

Mathematics Bootcamp

Properties of Preferences and Utility Functions

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Today

- Preference Relations, Indifference, and Utility
- Monotonicity and Local Nonsatiation
- Convexity and Quasiconcavity
- Continuity Through Contour Sets
- Continuous Utility Representation

Preference Relations

- A **preference relation** $\succeq$ on X is a subset of $X \times X$.
- $x \succeq y$ reads: x is *at least as good* as y (x is *weakly better* than y).
- From $\succeq$ we derive:
 - Strict preference: $x \succ y \iff (x \succeq y \text{ and not } y \succeq x)$.
 - Indifference: $x \sim y \iff (x \succeq y \text{ and } y \succeq x)$.

Properties of Indifference

Indifference Relation

If $\succeq$ is rational (i.e., complete and transitive), then the induced indifference relation $\sim$ is:

1. **Reflexive:** $x \sim x$ for all $x \in X$.
2. **Symmetric:** $x \sim y \implies y \sim x$.
3. **Transitive:** $x \sim y \wedge y \sim z \implies x \sim z$.

Equivalence Relation

An equivalence relation is a binary relation that is reflexive, symmetric and transitive.

Equivalence Classes

Equivalence Classes

An equivalence relation induces a **partition** of the set: mutually disjoint subsets whose union equals the entire set.

If $\succeq$ is rational, its induced indifference relation $\sim$

1. is an equivalence relation;
2. generates equivalence classes of indifferent elements.

For each $x \in X$, the equivalence class of x is denoted by

$$[x] = \{y \in X \mid x \sim y\}.$$

Equivalence Classes and Utility

Utility representation

A function $u : X \rightarrow \mathbb{R}$ *represents* $\succeq$ if

$$x \succeq y \iff u(x) \geq u(y).$$

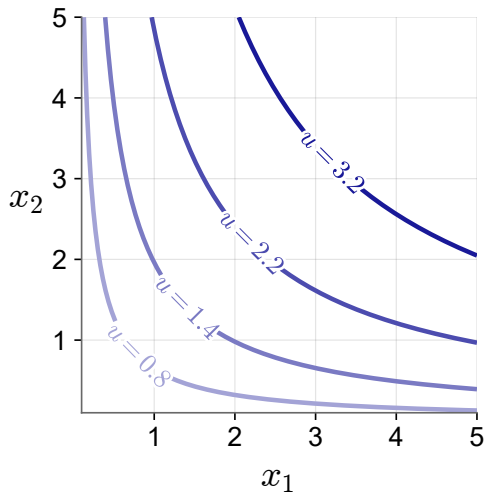
What the Representation Implies

If u represents $\succeq$, then

$$x \sim y \iff u(x) = u(y).$$

Every equivalence class maps to one utility level; utility is **ordinal**.

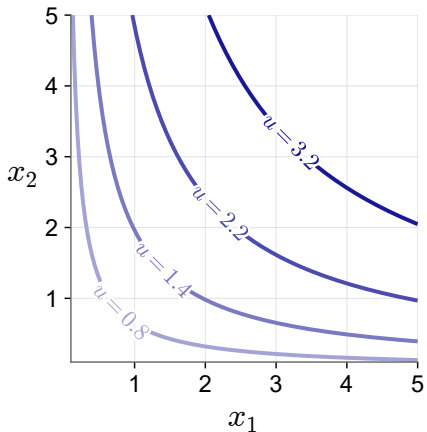
Indifference Curves



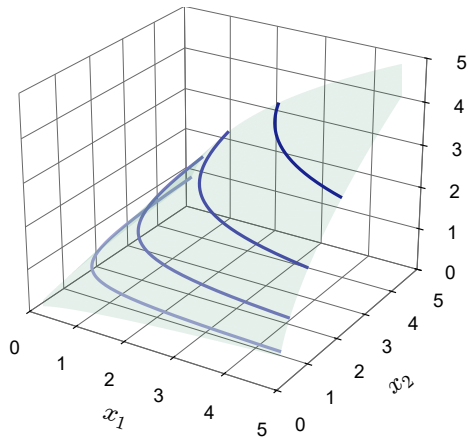
- Indifference curves are equivalence classes of X ; each maps to a single utility level in $\mathbb{R}$.
- Distinct equivalence classes are disjoint, so two indifference curves can never cross, or even touch.

Utility Representation

Indifference curves



Utility surface



Useful Properties of Preferences

- In addition to rationality, we assume more structure on preferences: more properties.
- Each property has implications for the utility representation and for the indifference curves.
- The best way to learn them is by practicing proofs, many times — that is what the problem sets are for.

Represent Commodities with Ordered Lists

2.B Commodities

The decision problem faced by the consumer in a market economy is to choose consumption levels of the various goods and services that are available for purchase in the market. We call these goods and services *commodities*. For simplicity, we assume that the number of commodities is finite and equal to L (indexed by $\ell = 1, \dots, L$).

As a general matter, a *commodity vector* (or *commodity bundle*) is a list of amounts of the different commodities,

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix},$$

and can be viewed as a point in $\mathbb{R}^L$, the *commodity space*.¹

We can use commodity vectors to represent an individual's consumption levels. The ℓ th entry of the commodity vector stands for the amount of commodity ℓ consumed. We then refer to the vector as a *consumption vector* or *consumption bundle*.

- $x \geq y$ iff $x_\ell \geq y_\ell$ for every ℓ ; and $x \gg y$ iff $x_\ell > y_\ell$ for every ℓ .
- So $x \geq y$ with $x \neq y$ means $x_\ell > y_\ell$ for *some* ℓ — weaker than $x \gg y$.

Monotonicity: More is Better

MWG 3.B.2: Monotonicity

Assume consuming more is always feasible: $x \in X$ and $y \geq x$ imply $y \in X$. The preference relation $\succeq$ on X is

monotone if $y \gg x$ implies $y \succ x$;

strongly monotone if $y \geq x$ and $y \neq x$ imply $y \succ x$.

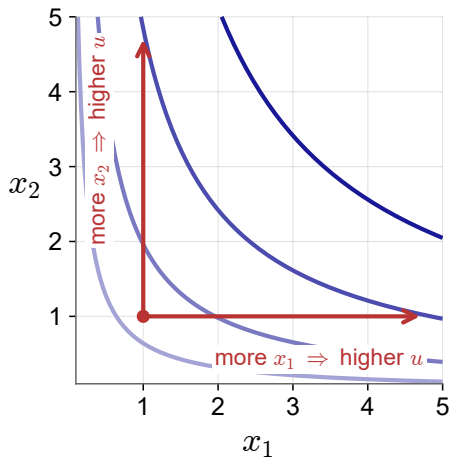
Proposition

If $u : X \rightarrow \mathbb{R}$ represents $\succeq$, then

$\succeq$ is monotone *iff* $u(y) > u(x)$ whenever $y \gg x$;

$\succeq$ is strongly monotone *iff* u is *strictly increasing*: $u(y) > u(x)$ whenever $y \geq x$ and $y \neq x$.

Monotonicity: More is Better



Domain: $X = \mathbb{R}_{++}^2$.

Local Nonsatiation: No Thick Indifference Curves

Definition: Local Nonsatiation

A preference relation $\succeq$ on X is *locally nonsatiated* if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ such that $\|x - y\| \leq \varepsilon$ and $y \succ x$.

$$\text{Euclidean distance: } \|x - y\| = \sqrt{\sum_{i=1}^L (x_i - y_i)^2}.$$

Local Nonsatiation: No Thick Indifference Curves

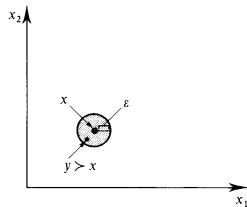


Figure 3.B.1

The test for local nonsatiation.

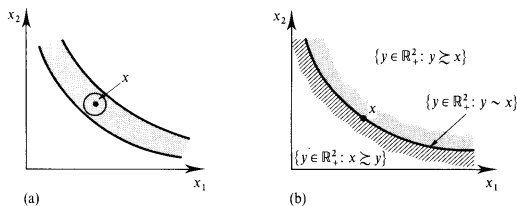


Figure 3.B.2

(a) A thick indifference set violates local nonsatiation.
(b) Preferences compatible with local nonsatiation.

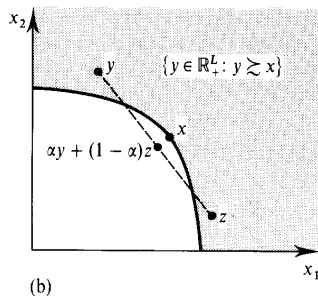
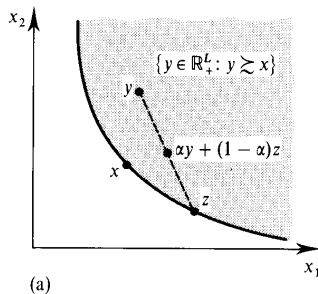
Monotonicity and Local Nonsatiation

Proposition

On $X = \mathbb{R}_+^L$, monotonicity implies local nonsatiation.

However, local nonsatiation does not imply monotonicity.

Convexity: Taste for Variety



Definition 3.B.4: The preference relation $\succsim$ on X is *convex* if for every $x \in X$, the upper contour set $\{y \in X: y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1 - \alpha)z \succsim x$ for any $\alpha \in [0, 1]$.

Figure 3.B.3(a) depicts a convex upper contour set; Figure 3.B.3(b) shows an upper contour set that is not convex.

Convexity: Taste for Variety

Convex Sets and Preferences

A set $X \subseteq \mathbb{R}^L$ is *convex* if $x, y \in X$ and $\alpha \in [0, 1]$ imply $\alpha x + (1 - \alpha)y \in X$.

Let X be convex and $U(x) := \{y \in X : y \succeq x\}$. Then $\succeq$ is *convex* if

$$y, z \in U(x), \alpha \in [0, 1] \implies \alpha y + (1 - \alpha)z \in U(x),$$

and *strictly convex* if that conclusion is $\succ x$ for $y \neq z, \alpha \in (0, 1)$.

Every upper contour set $U(x)$ is convex.

Level Sets and Quasiconcavity

Definitions

Let $X \subseteq \mathbb{R}^L$ be convex. A function $f : X \rightarrow \mathbb{R}$ is *quasiconcave* if, for $x, y \in X$ and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \geq \min\{f(x), f(y)\}.$$

It is *strictly quasiconcave* if, for $x \neq y$ and $\alpha \in (0, 1)$,

$$f(\alpha x + (1 - \alpha)y) > \min\{f(x), f(y)\}.$$

Equivalently. f is quasiconcave iff every upper level set $\{x \in X \mid f(x) \geq c\}$ is convex.

Convex Preferences and Quasiconcave Utility

Kreps 2.8(b)

Let X be convex and let $u : X \rightarrow \mathbb{R}$ represent $\succeq$. Then

$$\begin{aligned}\succeq \text{ is convex} &\iff u \text{ is quasiconcave,} \\ \succeq \text{ is strictly convex} &\iff u \text{ is strictly quasiconcave.}\end{aligned}$$

Properties of a Representation

Property of $\succsim$	Property of a representation u
Monotonicity	$u(y) > u(x)$ whenever $y \gg x$.
Strong monotonicity	u is strictly increasing.
Convexity	u is quasiconcave.
Strict convexity	u is strictly quasiconcave.

Continuity is different: that is Debreu's theorem.

Topology on the Choice Set

Relative Topology on $X \subseteq \mathbb{R}^L$

For $x \in X$ and $\varepsilon > 0$, the relative open ball is

$$B_\varepsilon^X(x) = \{y \in X : \|y - x\| < \varepsilon\}.$$

- A set $G \subseteq X$ is *open relative to X* if every $x \in G$ has some $\varepsilon > 0$ with $B_\varepsilon^X(x) \subseteq G$.
- A set $F \subseteq X$ is *closed relative to X* if $X \setminus F$ is open relative to X .

Day 2 defined open, closed and bounded in $\mathbb{R}^L$. The same definitions, with the balls cut down to X , give the relative notions.

Closed Sets Contain Their Limits

Sequential Test for Closed Sets

Let $F \subseteq X \subseteq \mathbb{R}^L$. The set F is closed relative to X if and only if

$$x_n \in F \text{ for every } n, \quad x_n \rightarrow x \in X \quad \implies \quad x \in F.$$

Lower and Upper Contour Sets

Contour Sets at x

$$U(x) := \{y \in X \mid y \succeq x\} \quad L(x) := \{y \in X \mid x \succeq y\}.$$

The indifference set through x is always

$$I(x) := \{y \in X : y \sim x\} = U(x) \cap L(x).$$

Weak and Strict Sections

If $\succeq$ is complete, then

$$\{y \in X : y \succ x\} = X \setminus L(x), \quad \{y \in X : x \succ y\} = X \setminus U(x).$$

The indifference set is both the intersection of the two contour sets and their shared boundary. If the contour sets are closed, that boundary is included: no gaps in the indifference curves.

Lexicographic Preferences

Example 3.C.1: The Lexicographic Preference Relation

Let $X = \mathbb{R}_+^2$. Define $x \succeq y$ if either

$$x_1 > y_1, \quad \text{or} \quad x_1 = y_1 \text{ and } x_2 \geq y_2.$$

It is complete, transitive, strongly monotone, and strictly convex — yet no utility function represents it.

Its contour sets are not closed: $(1 + \frac{1}{n}, 0) \succ (1, 1)$ for every n , yet the limit $(1, 0)$ falls strictly below $(1, 1)$. The upper contour set of $(1, 1)$ loses its boundary point.

Continuity: Closed Contour Sets

Continuity

Let $X \subseteq \mathbb{R}^L$ and let $\succeq$ be complete. Then $\succeq$ is *continuous* if, for every $x \in X$,

$$U(x) \text{ and } L(x) \text{ are closed relative to } X.$$

Exactly the property the lexicographic relation just failed. No boundary point of a contour set escapes: no gaps.

Continuity: Equivalent Tests

Kreps 1.14: Equivalent Characterizations

For complete, transitive $\succeq$, continuity is equivalent to each of:

1. $U(x)$ and $L(x)$ are closed relative to X , for every x .
2. $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open relative to X , for every x .
3. If $x \succ y$, there is $\varepsilon > 0$ with $x' \succ y'$ whenever $\|x' - x\| < \varepsilon$ and $\|y' - y\| < \varepsilon$.

MWG instead defines continuity through limits of rankings (3.C.1):

$$(x_n, y_n) \rightarrow (x, y), \quad x_n \succeq y_n \text{ for every } n \quad \implies \quad x \succeq y,$$

equivalent to all three. Kreps takes (3); we take (1), the one the pictures use.

Continuous Representations and Closed Contours

Closed Level Sets

If $f : X \rightarrow \mathbb{R}$ is continuous, then for every $c \in \mathbb{R}$ the sets

$$f^{-1}([c, \infty)) \quad \text{and} \quad f^{-1}((-\infty, c])$$

are closed relative to X .

Closed contour sets are the definition, so a continuous representation gives continuous preferences.

Debreu's Representation Theorem

Kreps 2.9: Debreu's Theorem

Let $X = \mathbb{R}_+^L$. If $\succeq$ is complete, transitive, and continuous on X , then there exists a *continuous* utility function $u : X \rightarrow \mathbb{R}$ such that

$$x \succeq y \iff u(x) \geq u(y).$$

- It gives *at least one* continuous representation — not every representation is continuous.

Three Assumptions, Three Conclusions

Assumptions	Conclusion
Nonempty compact D ; continuous u	$\arg \max_{x \in D} u(x)$ is nonempty.
Convex D ; strictly quasiconcave u	There is at most one maximizer.
Local nonsatiation; $D = B(p, I)$ with $p \gg 0$	Every optimum satisfies $p \cdot x = I$.

Day 4 takes each in turn.